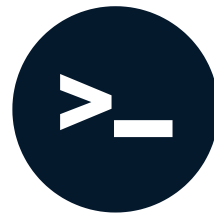


Intermediate YAML

CI/CD FOR MACHINE LEARNING

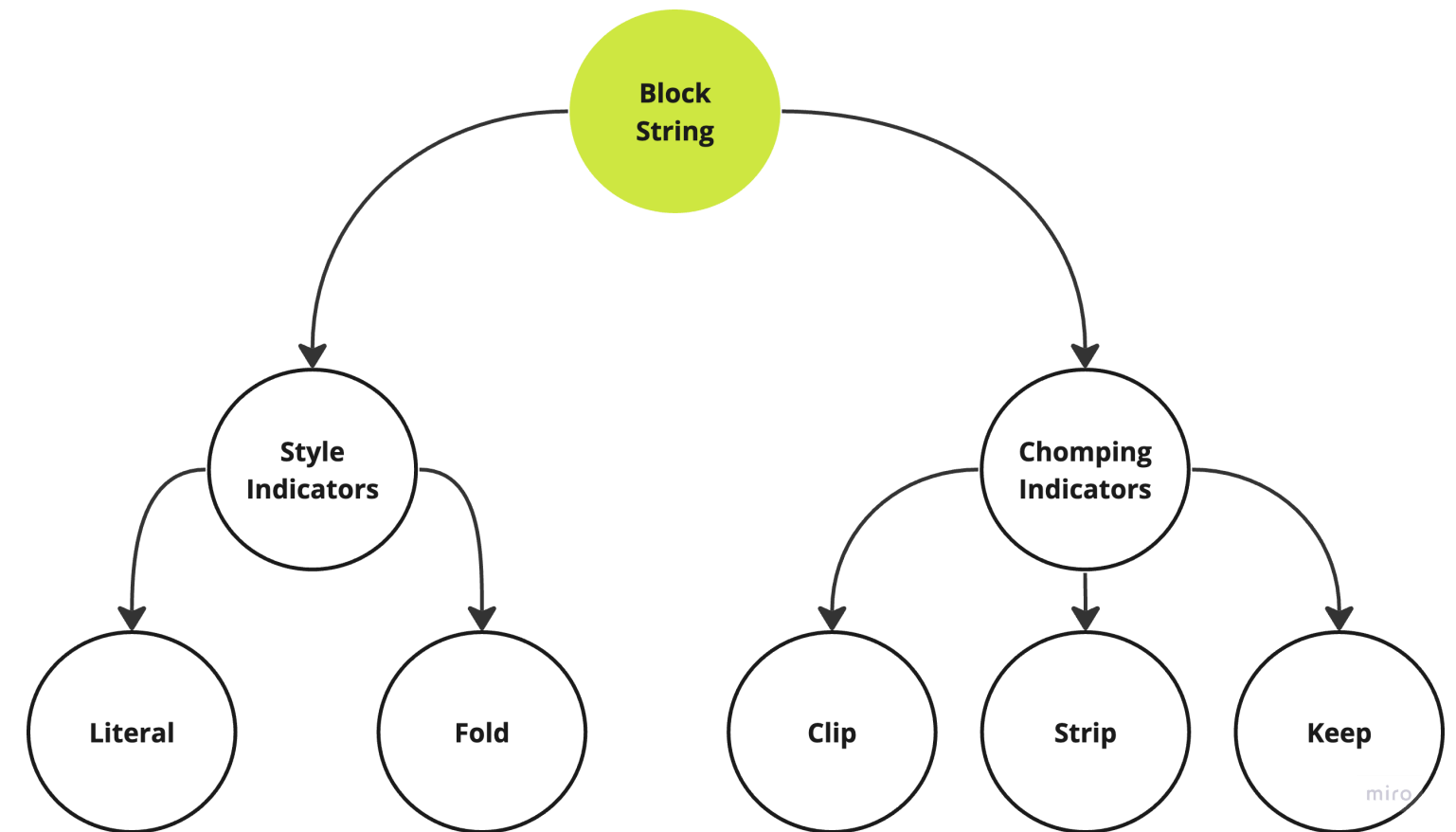


Ravi Bhaduria

Machine Learning Engineer

Multiline strings: Block scalar format

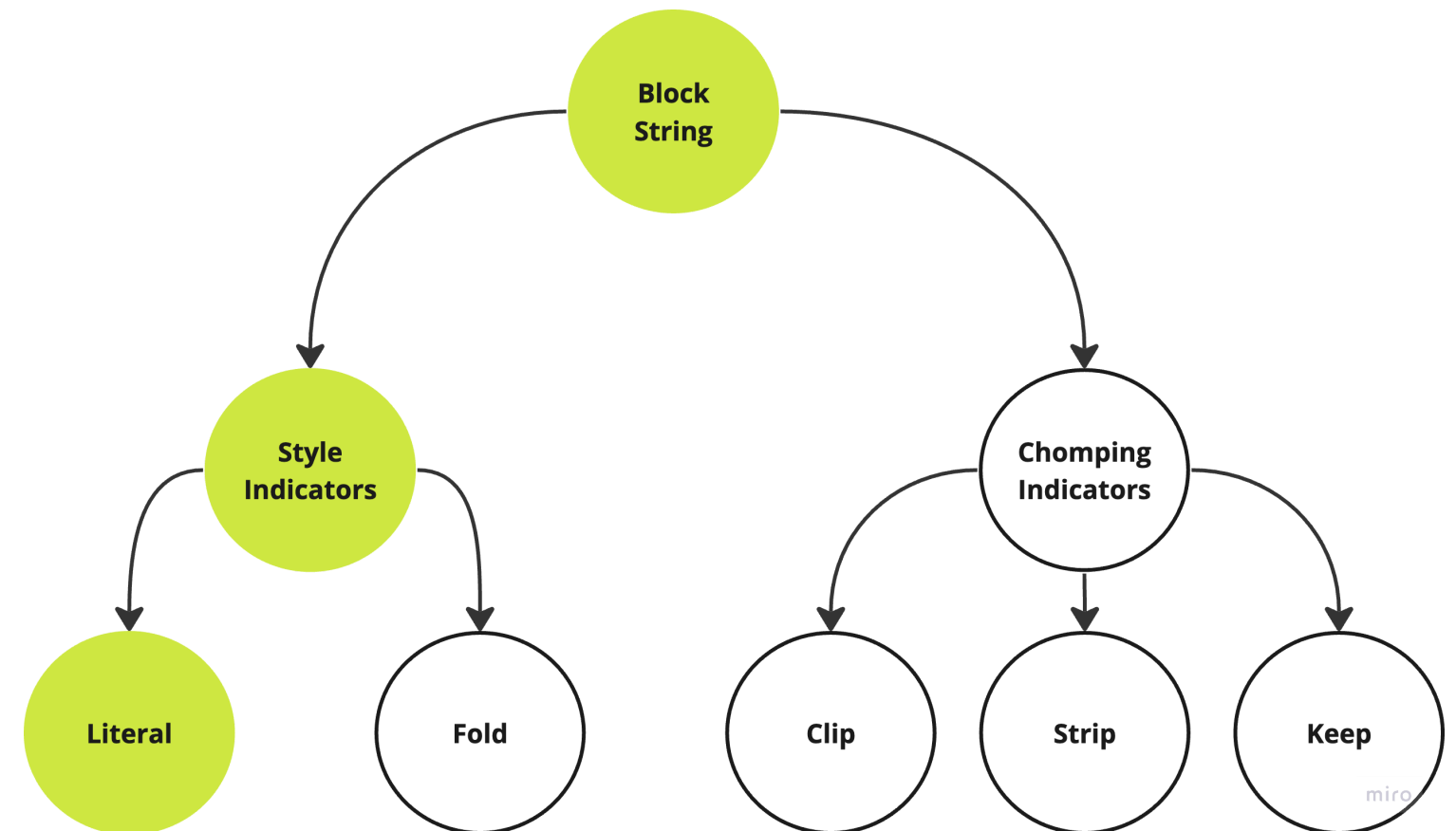
- Represent multiline strings or blocks of text
 - Code snippets (shell commands)
 - Paragraphs (logs)
 - Configuration files



Style indicators - literal

- Literal style (`|`): preserves line break and indentation
- Useful for writing shell commands

```
commands: |  
    echo "Running python script..."  
    python3 script.py
```



Literal style example

YAML

example: |
|

```
..Several lines of text,\n..with some "quotes" of various 'types',\n..and also a blank line:\n..\n..and some text with\n....extra indentation\n..on the next line,\n..plus another line at the end.\n..\n..\n
```

Result

```
Several lines of text,\nwith some "quotes" of various 'types',\nand also a blank line:\n\nand some text with\n    extra indentation\non the next line,\nplus another line at the end.
```

miro

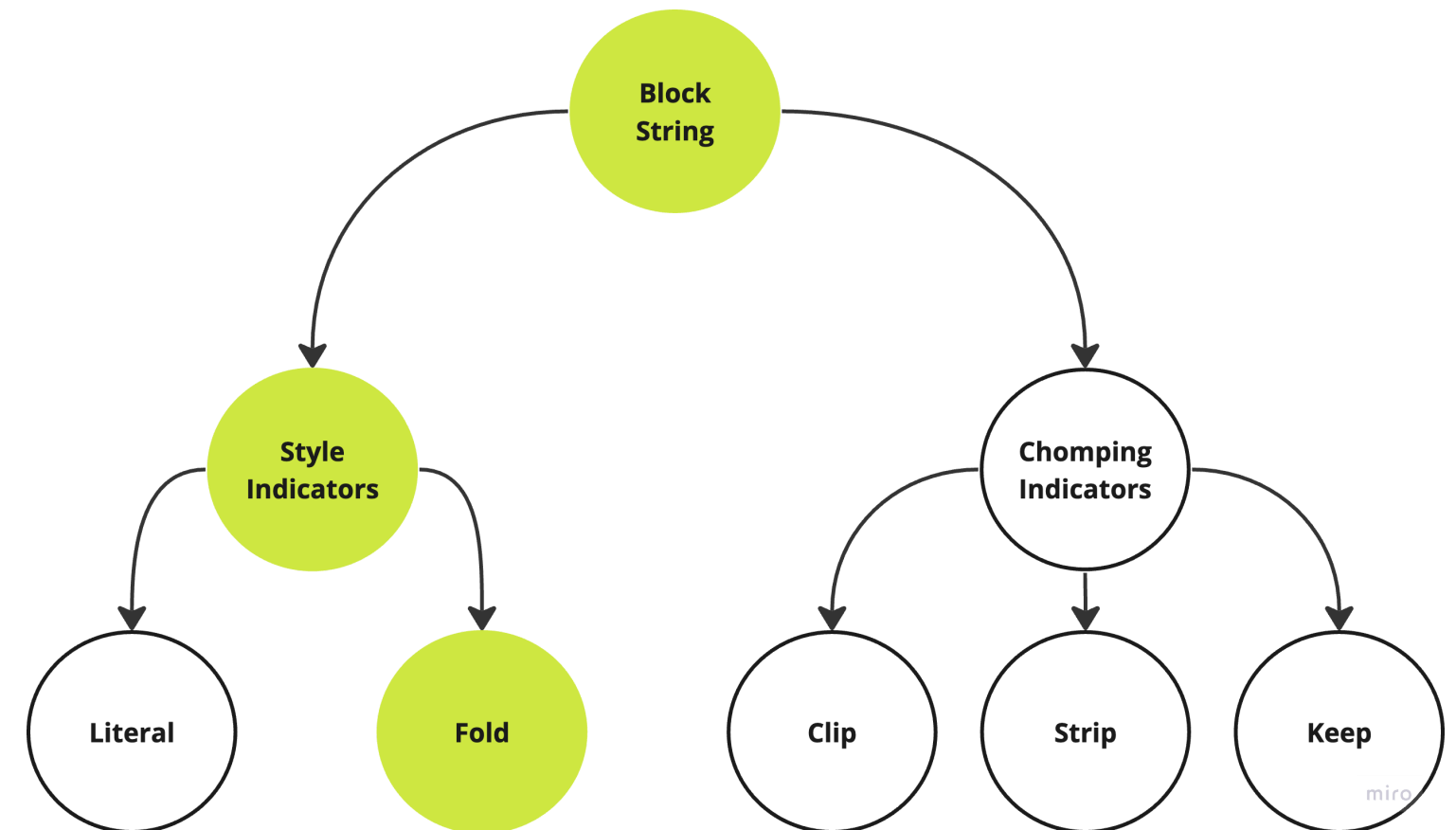
¹ <https://yaml-multiline.info/>

Style indicators - fold

- Fold style (>): removes line breaks
- Useful for writing wrapped messages like paragraphs

message: >

```
Successfully executed the code!  
All processes completed  
without any errors.  
The results have been saved  
in the designated  
output directory.
```



Fold style example

YAML

example: >\n

```
..Several lines of text,\n..with some "quotes" of various 'types',\n..and also a blank line:\n..\n..and some text with\n....extra indentation\n..on the next line,\n..plus another line at the end.\n..\n..\n
```

Result

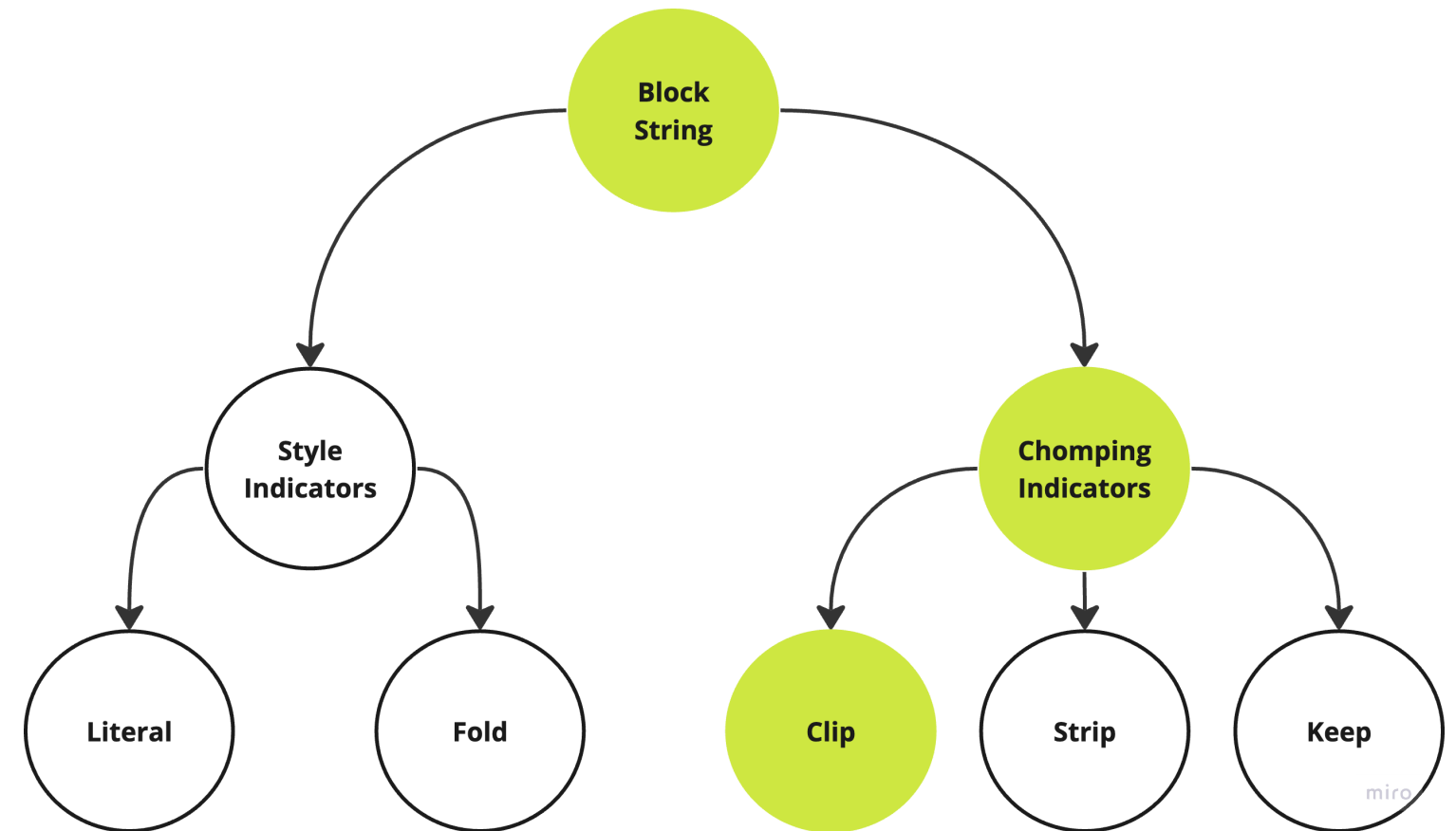
Several lines of text, with some "quotes" of various 'types', and also a blank line:
and some text with
 extra indentation
on the next line, plus another line at the end.

miro

¹ <https://yaml-multiline.info/>

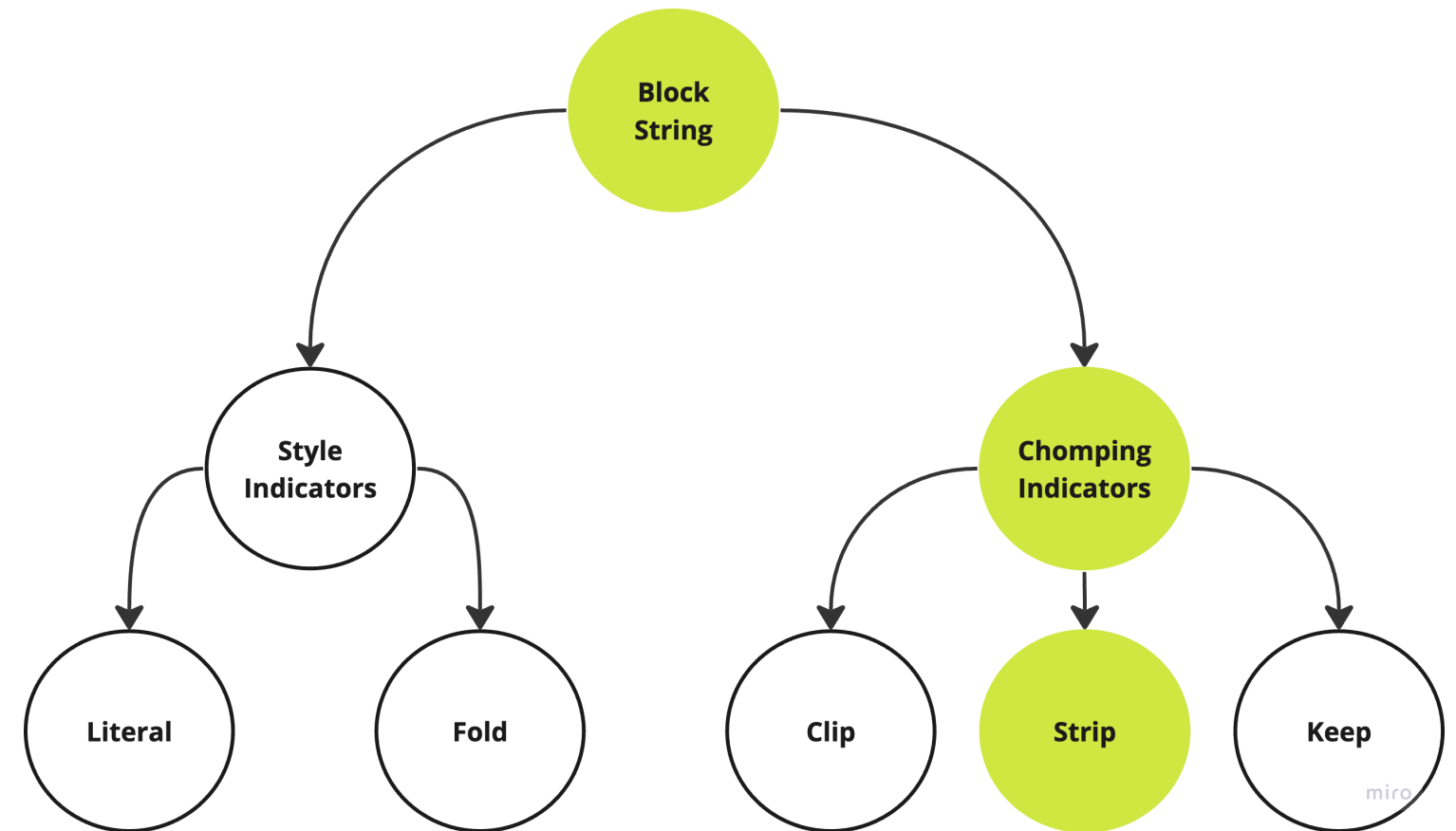
Chomping indicators

- Control the behavior of newlines at the *end* of the string
- Added after the style indicators
- **clip**: default mode, single newline at the end
 - No additional symbol needed for representation



Chomping indicators - strip

- `strip (-)`: removes all newlines at the end



Strip chomping example

YAML

example: |-\n

```
..Several lines of text,\n..with some "quotes" of various 'types',\n..and also a blank line:\n..\n..and some text with\n....extra indentation\n..on the next line,\n..plus another line at the end.\n\n\n
```

Result

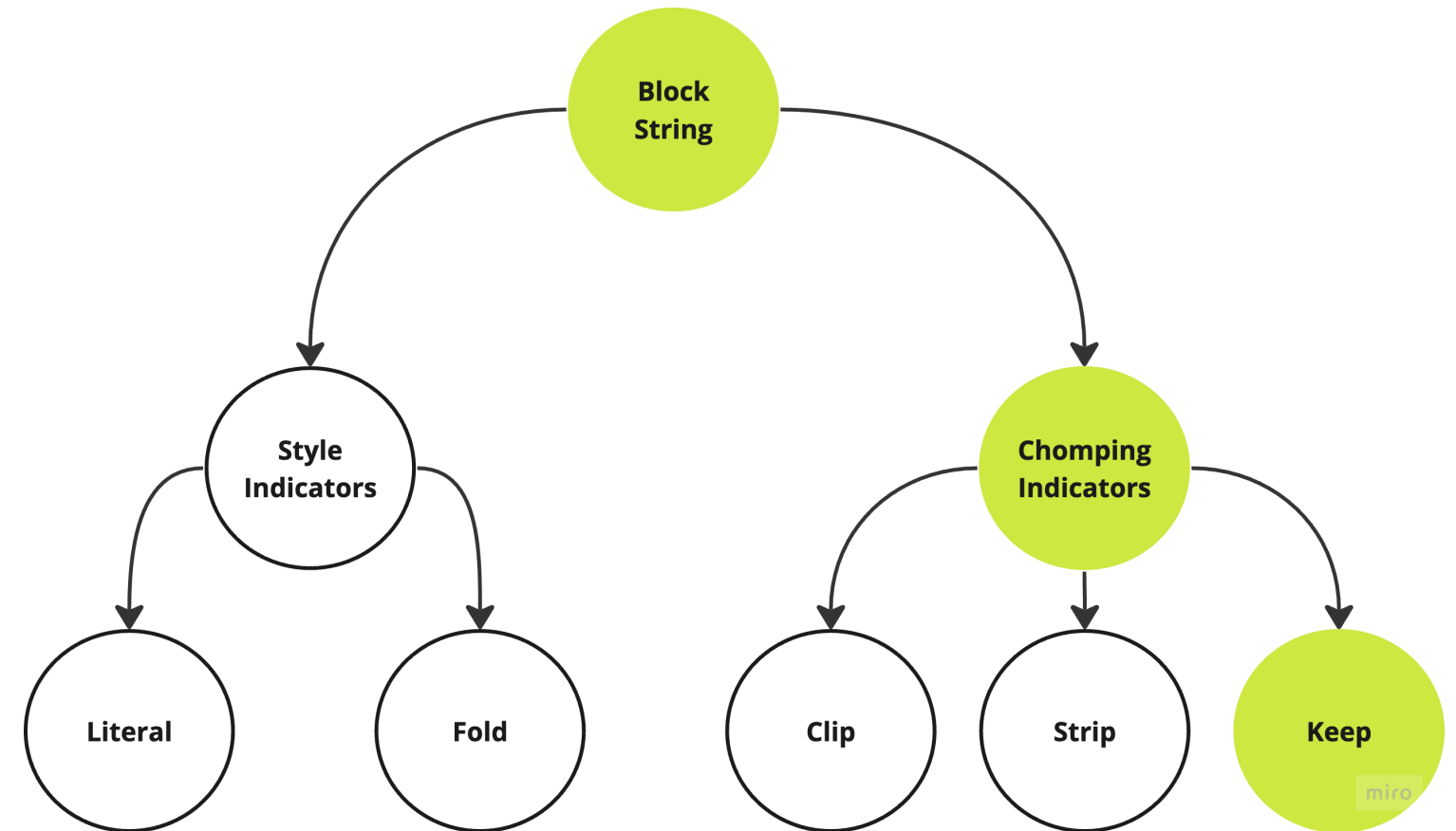
```
Several lines of text,\nwith some "quotes" of various 'types',\nand also a blank line:\n\nand some text with\n    extra indentation\non the next line,\nplus another line at the end.
```

miro

¹ <https://yaml-multiline.info/>

Chomping indicators - keep

- `keep (+)`: retains all newlines at the end



Keep chomping example

YAML

```
example: |+  
..Several lines of text,  
..with some "quotes" of various 'types',  
..and also a blank line:  
..  
..and some text with  
....extra indentation  
..on the next line,  
..plus another line at the end.  
..  
..
```

Result

```
Several lines of text,  
with some "quotes" of various 'types',  
and also a blank line:  
  
and some text with  
    extra indentation  
on the next line,  
plus another line at the end.  
  

```

miro

¹ <https://yaml-multiline.info/>

Dynamic value injection

- Expressions allow parsers to dynamically substitute values
- Usage:
 - Environment variables
 - References to other parts of YAML
- Typically represented as `${{ ... }}`

```
database:  
  host: ${ config.database.host }  
  port: ${ config.database.port }  
  username: ${ config.database.username }  
  password: ${ config.database.password }
```

Multi-document YAML

- Multiple separate YAML documents within a single file
- Each document contains valid YAML content
- Easy to organize by related documents
- Separate documents by three dashes ---

```
---  
- name: John  
  age: 30  
  
---  
- name: Jane  
  age: 28  
  
---  
- name: Bob  
  age: 35  
  occupation: Developer
```

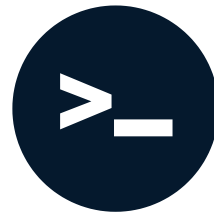
Summary

Name	YAML Symbol
Literal Style Indicator	
Fold Style Indicator	>
Strip Chomping Indicator	-
Keep Chomping Indicator	+
Expressions	<code>\${{ ... }}</code>
Multi-document YAML	---

Let's practice!
CI/CD FOR MACHINE LEARNING

Setting a basic CI pipeline

CI/CD FOR MACHINE LEARNING



Ravi Bhaduria
Machine Learning Engineer

Anatomy of GitHub Actions workflow

```
# Name of the workflow
name: CI

# Events that trigger workflow
on:
  push:
    branches: [ "main" ]
```

Anatomy of GitHub Actions workflow

```
# Define jobs and runners
jobs:
  build:
    runs-on: ubuntu-latest
# Define steps
steps:
  - name: Run a multi-line script
    run: |
      echo Hello, world!
      echo Add other actions to build,
      echo test, and deploy your project.
```

Anatomy of GitHub Actions workflow

```
name: CI

on:
  push:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Run a multi-line script
        run: |
          echo Hello, world!
          echo Add other actions to build,
          echo test, and deploy your project.
```

Create a GitHub repository

- <https://github.com/new>

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

Owner *

 rbhadauria29 ▾

Repository name *

ci-cd-for-ml-demo

✓ ci-cd-for-ml-demo is available.

Great repository names are short and memorable. Need inspiration? How about [jubilant-doodle](#) ?

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:



Add a README file

This is where you can write a long description for your project. [Learn more about READMEs](#).

Add .gitignore

.gitignore template: Python ▾

Choose which files not to track from a list of templates. [Learn more about ignoring files](#).

Choose a license

License: MIT License ▾

A license tells others what they can and can't do with your code. [Learn more about licenses](#).

This will set  main as the default branch. Change the default name in your [settings](#).

 You are creating a public repository in your personal account.

Create repository

Set up GitHub Actions

- <https://github.com/rbhadauria29/ci-cd-for-ml-demo/settings/actions>

Actions permissions

☒ **Allow all actions and reusable workflows**

Any action or reusable workflow can be used, regardless of who authored it or where it is defined.

☐ **Disable actions**

The Actions tab is hidden and no workflows can run.

☐ **Allow rbhadauria29 actions and reusable workflows**

Any action or reusable workflow defined in a repository within rbhadauria29 can be used.

☐ **Allow rbhadauria29, and select non-rbhadauria29, actions and reusable workflows**

Any action or reusable workflow that matches the specified criteria, plus those defined in a repository within rbhadauria29, can be used. [Learn more about allowing specific actions and reusable workflows to run.](#)

Save

Setup a "Hello, World!" GitHub Action

rbhadauria29 / ci-cd-for-ml-demo

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

ci-cd-for-ml-demo Public

 Pin

 Unwatch 1

 main

 1 branch

 0 tags

Go to file

Add file

<> Code

rbhadauria29 Initial commit

57fbeed 50 minutes ago 1 commit

 .gitignore	Initial commit	50 minutes ago
 LICENSE	Initial commit	50 minutes ago
 README.md	Initial commit	50 minutes ago

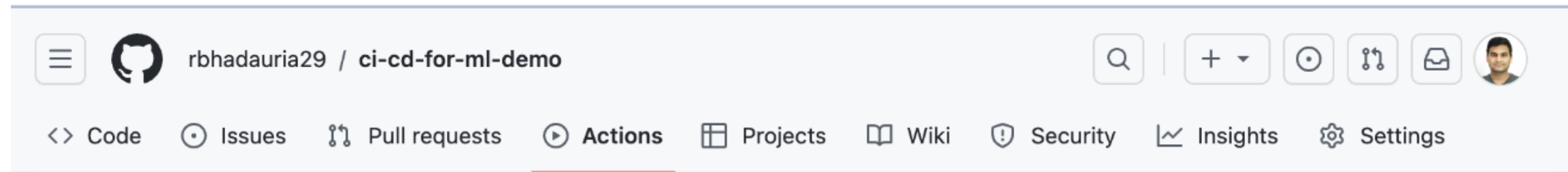
README.md



ci-cd-for-ml-demo

miro

Setup a "Hello, World!" GitHub Action



Get started with GitHub Actions

Build, test, and deploy your code. Make code reviews, branch management, and issue triaging work the way you want. Select a workflow to get started.

Skip this and [set up a workflow yourself](#) →

Suggested for this repository

Simple workflow

By GitHub

Start with a file with the minimum necessary structure.

Configure

miro

Setup a "Hello, World!" GitHub Action

ci-cd-for-ml-demo / .github / workflows / hello-world.yml in main

Edit Preview

Spaces 2 No wrap

```
1 # This is a basic workflow to help you get started with Actions
2
3 # Name of the workflow |
4 name: CI
5
6 # Controls when the workflow will run
7 on:
8   # Triggers the workflow on push events but only for the "main" branch
9   push:
10     branches: [ "main" ]
11
12 # A workflow run is made up of one or more jobs that can run sequentially or in parallel
13 jobs:
14   # This workflow contains a single job called "build"
15   build:
16     # The type of runner that the job will run on
17     runs-on: ubuntu-latest
18
19     # Steps represent a sequence of tasks that will be executed as part of the job
20     steps:
21
22     # Runs a set of commands using the runners shell
23     - name: Run a multi-line script
24       run: |
25         echo Hello, world!
26         echo Add other actions to build,
27         echo test, and deploy your project.
28
```

Cancel changes Commit changes...

Marketplace Documentation

Search Marketplace for Actions

Featured Actions

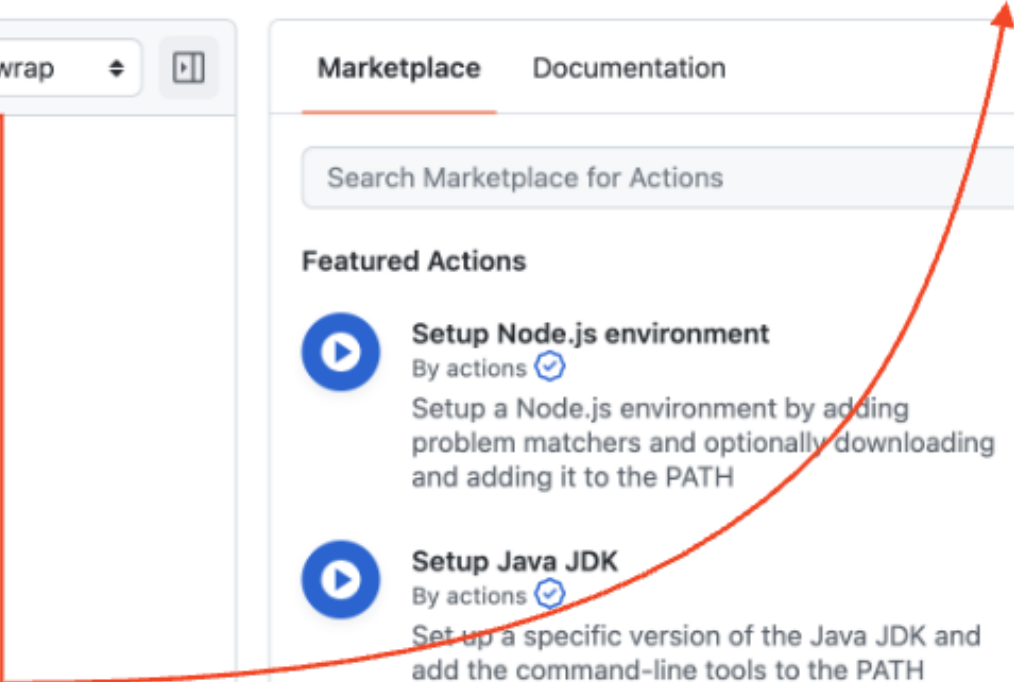
Setup Node.js environment 3.1k

Setup Java JDK 1.2k

Setup Go environment 1.1k

Close Stale Issues 1k

First interaction 633



Setup a "Hello, World!" GitHub Action

Commit changes

Commit message

Hello, World! GA pipeline

Extended description

Add an optional extended description..

☒ Commit directly to the main branch

☐ Create a **new branch** for this commit and start a pull request
[Learn more about pull requests](#)

Cancel

Commit changes

Inspect the GitHub Actions Run

 rbhadauria29 Hello, World! GA pipeline		✓ e49adbd 12 minutes ago	🕒 5 commits
 .github/workflows	Hello, World! GA pipeline		12 minutes ago
 .gitignore	Initial commit		yesterday
 LICENSE	Initial commit		yesterday
 README.md	Initial commit		yesterday

Inspect the GitHub Actions Run

All workflows

Showing runs from all workflows

Q Filter workflow runs

1 workflow run

Event ▼ Status ▼ Branch ▼ Actor ▼

✓ Hello, World! GA pipeline

CI #4: Commit e49adbd pushed by rbhadauria29

main

11 minutes ago

12s

...

Triggered via push 1 hour ago

Status

Total duration

Artifacts

rbhadauria29 pushed

e49adbd

main

Success

12s

—

hello-world.yml

on: push

✓ build

1s

miro

Inspecting the Logs

← CI

✓ Hello, World! GA pipeline #4

Re-run all jobs

...

Summary

Jobs

build

Run details

Usage

Workflow file

build

succeeded 1 hour ago in 1s

Search logs

↺

⚙

Set up job

0s

1 Current runner version: '2.307.1'

2 ▶ Operating System

6 ▶ Runner Image

11 ▶ Runner Image Provisioner

13 ▶ GITHUB_TOKEN Permissions

17 Secret source: Actions

18 Prepare workflow directory

19 Prepare all required actions

20 Complete job name: build

Run a multi-line script

1s

1 ▶ Run echo Hello, world!

6 Hello, world!

7 Add other actions to build,

8 test, and deploy your project.

Complete job

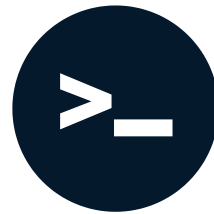
0s

1 Cleaning up orphan processes

Let's practice!
CI/CD FOR MACHINE LEARNING

Running repository code

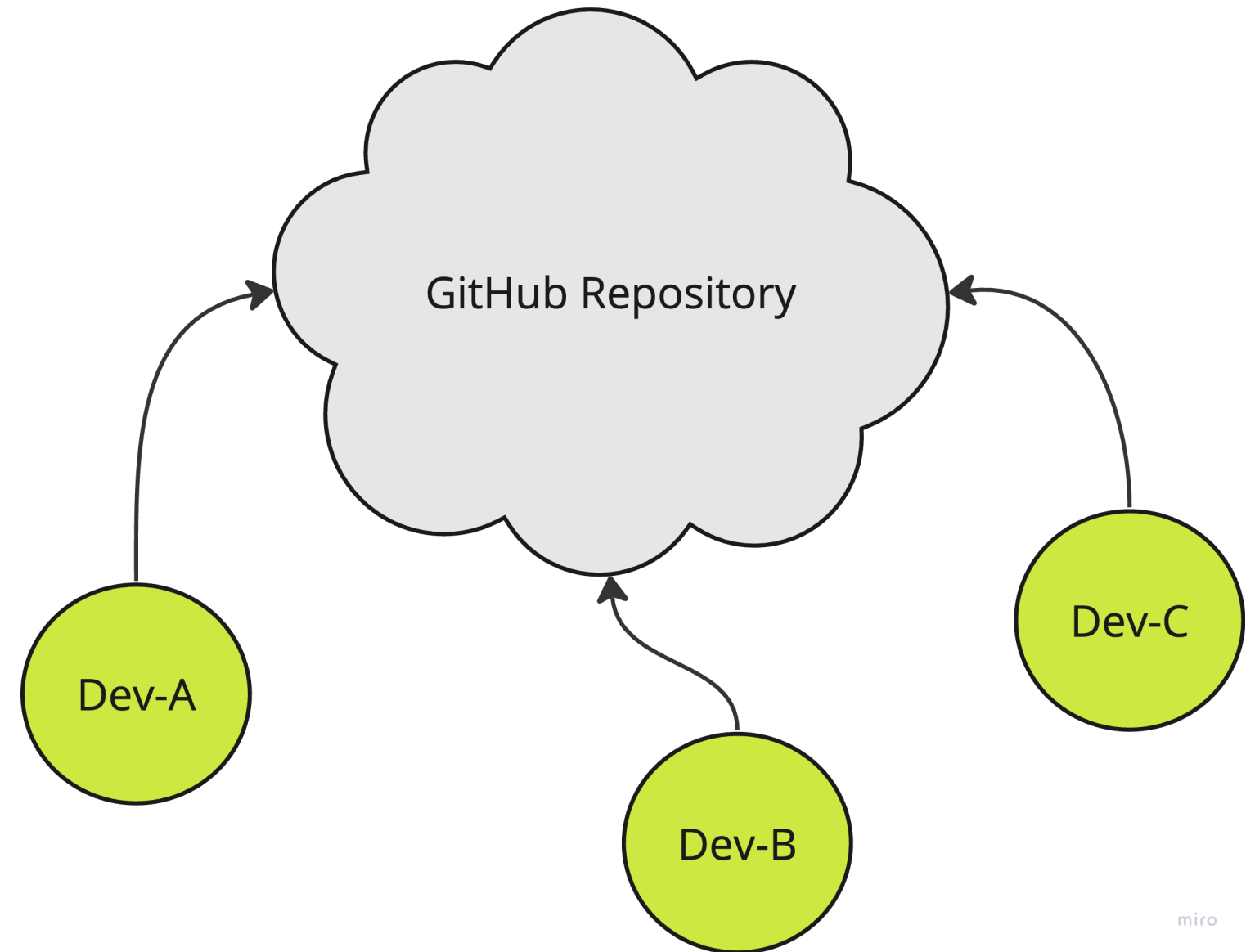
CI/CD FOR MACHINE LEARNING



Ravi Bhaduria
Machine Learning Engineer

Shared repository model

- Multiple collaborators work on same repository
- Feature/Topic branches are created for related work
 - Pull requests are opened
 - Code is reviewed and changes are suggested
 - Branch code is merged into `main` branch
- CI/CD enabled checks improve code quality



Create a feature branch

 **ci-cd-for-ml-demo** Public

 Pin

 Unwatch 1 ▾

 Fork 0 ▾

 Star 0 ▾

 main ▾

 **1 branch**

 **0 tags**

Go to file

Add file ▾

 **Code** ▾

About



 Search branches...

Overview

Yours

Active

Stale

All branches

New branch

Default branch



main

 Updated 9 hours ago by rbhadauria29

✓

Default



 datacamp

CI/CD FOR MACHINE LEARNING

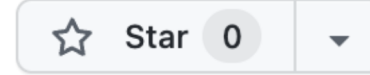
Create a feature branch

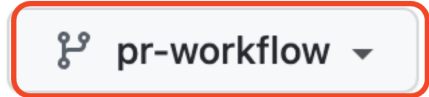
 ci-cd-for-ml-demo Public

 Pin

 Unwatch 1 ▾

 Fork 0 ▾

 Star 0 ▾

 pr-workflow ▾

 2 branches  0 tags

Go to file

Add file ▾

 Code ▾

About



This branch is up to date with main.

 Contribute ▾

No description, website, or topics provided.

miro

 README

Add repository code

Save as `hello_world.py` in the branch

```
import datetime

def main():
    print("Hello, World!")

    # Get the current time and print it
    current_time = datetime.datetime.now()
    print("Current time:", current_time)

if __name__ == "__main__":
    main()
```

Configure workflow event

- Specify target branch for pull request event

```
on:  
  pull_request:  
    branches: [ "main" ]
```

Actions syntax

steps:

- name: My Action

- uses: <org_or_user_name>/<reponame>@<version_number>

- with:

- action-argument: value

- Action repository https://github.com/<org_or_user_name>/<reponame>
 - E.g. <https://github.com/actions/checkout>
- Arguments can be provided using *with* key
- GitHub marketplace actions at <https://github.com/actions>

Configure workflow steps and actions

- Checking out repository in job step

steps:

- name: Checkout
uses: actions/checkout@v3

- Setting up python environment

steps:

- name: Setup Python
uses: actions/setup-python@v4
with:
python-version: 3.9

Putting it together

- Workflow triggers on pull request to `main`
- `build` job contains three steps
 - `Checkout` checks out the repository
 - `Setup Python` sets up the Python environment
 - `Run Python script` runs the Python file

```
name: PR

on:
  pull_request:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout
        uses: actions/checkout@v3
      - name: Setup Python
        uses: actions/setup-python@v4
        with:
          python-version: 3.9
      - name: Run Python script
        run: |
          echo hello_world.py
          python hello_world.py
```

Create PR and trigger workflow

testing PR workflow #1

Open

rbhadauria29 wants to merge 2 commits into `main` from `pr-workflow`

Conversation 0

Commits 2

Checks 0

Files changed 2

+46 -0

rbhadauria29 commented now

Owner

No description provided.

rbhadauria29 added 2 commits 47 minutes ago

Create `hello_world.py`

Verified b93182b

Create `pr.yml`

Verified bcc63f6

Add more commits by pushing to the `pr-workflow` branch on `rbhadauria29/ci-cd-for-ml-demo`.

Some checks haven't completed yet

1 queued check

PR / build (pull_request)

Queued — Waiting to run this check...

Details

This branch has no conflicts with the base branch

Merging can be performed automatically.

Merge pull request

You can also [open this in GitHub Desktop](#) or view [command line instructions](#).

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

 datacamp

CI/CD FOR MACHINE LEARNING

Inspect workflow logs

rbhadauria29 / ci-cd-for-ml-demo

Q

Type ↵ to search

>_

+

+

+

+

+

+

<> Code

Issues

Pull requests 1

Actions

Projects

Wiki

Security

Insights

Settings

← Back to pull request #1

✓ testing PR workflow #1

Re-run all jobs

...

Summary

Jobs

✓ build

Run details

Usage

Workflow file

build

search logs

refresh

settings

succeeded 16 hours ago in 5s

> ✓ Set up job

2s

> ✓ Checkout

1s

> ✓ Setup Python

0s

▼ ✓ Run Python script

0s

1 ▶ Run echo hello_world.py

12 hello_world.py

13 Hello, World!

14 Current time: 2023-08-04 01:11:24.991641

> ✓ Post Setup Python

0s

> ✓ Post Checkout

0s

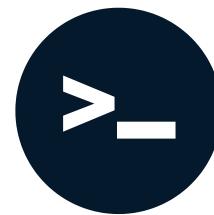
> ✓ Complete job

0s

Let's practice!
CI/CD FOR MACHINE LEARNING

Environment Variables and Secrets

CI/CD FOR MACHINE LEARNING



Ravi Bhaduria
Machine Learning Engineer

Contexts

- Access information about predefined variables and data
 - workflow runs
 - variables
 - runner environments
 - jobs and steps
- Access contexts using the expression syntax `${{ context.XXX }}`
- Contexts used in this course
 - `github` : information about the workflow run
 - `env` : variables set in the workflow
 - `secrets` : names and values that are available to workflow
 - `job` : info about the current job
 - `runner` : info about the machine

¹ <https://docs.github.com/en/actions/learn-github-actions/contexts>

Variables

- Store non-sensitive information in plain text
 - compiler flags, usernames, file paths
- Declared as value for `env` key
- Global/local scope is controlled by the level where defined
- Accessed from the `env` context as `{{ env.ENV_VAR }}`

```
name: Greeting on variable day
# Global env
env:
  Greeting: Hello
jobs:
  greeting_job:
    runs-on: ubuntu-latest
    # Local env: scoped to greeting_job
    env:
      First_Name: Ravi
    steps:
      - run: |
          echo "{{ env.Greeting }}" \
            "{{ env.First_Name }}"
```

Secrets

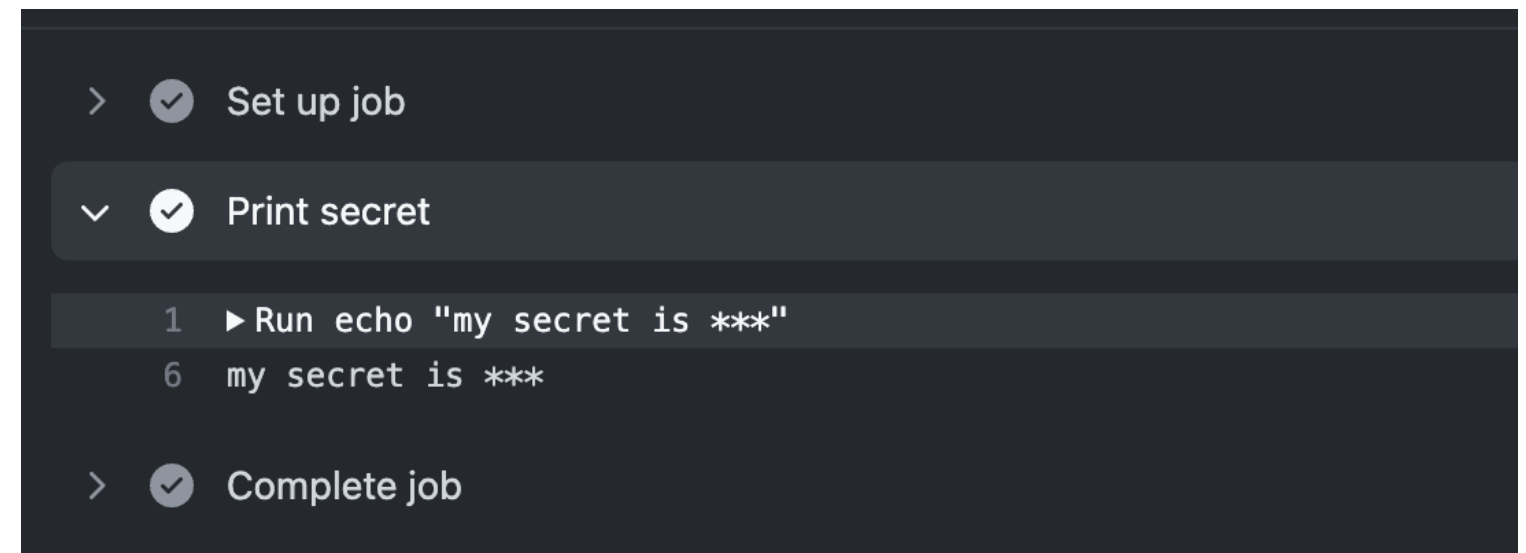
- Store sensitive information in encrypted manner
 - passwords, API keys
- Access values via `secrets` context
 - `{{ secrets.SuperSecret }}`
- Can store them as input or environment variable

```
steps:
  - name: Hello world action
    env: # Set the secret as an env var
      super_secret: {{ secrets.SuperSecret }}
    with: # Or as an input
      super_secret: {{ secrets.SuperSecret }}
```

- Printing secret

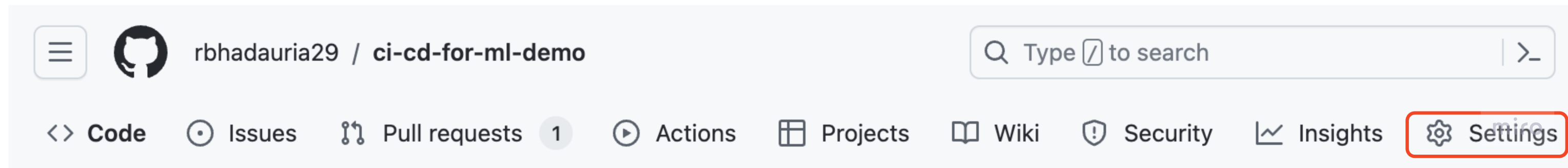
```
steps:
  - name: Print secret
    run: |
      echo "my secret is \
      {{ secrets.SuperSecret }}"
```

- Output



The screenshot shows a CI/CD pipeline interface with a dark background. It displays a list of steps: 'Set up job' (checked), 'Print secret' (checked and expanded), and 'Complete job' (checked). The 'Print secret' step is expanded to show two lines of output: '1 ▶ Run echo "my secret is ***"' and '6 my secret is ***'.

Setting secrets



- Go to Security > Secrets and Variables > Actions

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for **non-sensitive** data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.



Setting secrets

[Actions secrets](#) / New secret

Name *

SuperSecret

Secret *

secretValue

Add secret

GITHUB_TOKEN secret

- Built in secret provided by GitHub Actions
- Used to perform workflow actions
 - Cloning the repository and fetching code
 - Opening and closing issues and pull requests
 - Commenting on issues and pull requests
- Automatically available in every GitHub Actions workflow
 - Accessed via `${{ secrets.GITHUB_TOKEN }}`
- Permissions can be tuned to the right degree

Example: commenting on a pull request

- Grant permissions to write comments in PR

```
permissions:  
  pull-requests: write
```

- Use GITHUB_TOKEN to authorize

```
permissions:  
  pull-requests: write  
steps:  
  - name: Comment PR  
    uses: thollander/actions-comment-pull-request@v2  
    with:  
      GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}  
      message: |  
        Hello world ! :wave:
```



rbhadauria29 commented 6 minutes ago

Owner ...

No description provided.



Create pr-comment.yml

Verified ✓ 3988e7c



github-actions bot commented 6 minutes ago

...

Hello world ! 🖐️



¹ <https://gist.github.com/rbhadauria29/6d7fc51944b4fb48425c3c307fec77c6>

Let's practice!
CI/CD FOR MACHINE LEARNING